



TECHNICAL NOTE — DESIGN AND SIMULATION (SO FAR)

Low-Cost Active Control High Power Rocket

Starscream is Angel's first active control rocket and his personal third rocket, designed to reach a 1.2km apogee with a Student-Research and Developed (SRAD) Solid Rocket Motor (SRM) and control the rocket trajectory during Coasting flight.

AUTHOR	DATE	STATUS
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This brief technical report shows the results of Starscream, its flight computer, its SRM and its linear dynamics analysis.



01 Introduction

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All experiments were conducted at the Vultur Rocketry static test facility. Chamber pressure was measured with a Honeywell MLH series transducer (0–1000 psi). Thrust was recorded via a single-axis load cell, sampled at 1 kHz.

NOTE

Burn rate data presented here corresponds to ambient conditions of 2240 m ASL, T = 18 °C. Altitude corrections are applied in Section 3.

02 Propulsion

2.1 BURN RATE LAW

In addition to serving as chief engineer, Angel was directly involved in the design, manufacturing, and testing of the Solid Rocket Motor used in this mission. The VR-01 motor used a potassium nitrate / sorbitol (KNSB) propellant grain in a BATES configuration.

$$r = a \cdot P_n^n$$

(Eq. 1)

- r**propellant surface regression rate [mm/s]
- a**burn rate coefficient [mm/s · MPa⁻ⁿ]
- P_n**chamber pressure [MPa]
- n**pressure exponent [dimensionless]

2.2 THRUST AND SPECIFIC IMPULSE

$$F = I_{sp} \cdot \dot{m} \cdot g_0 + (P_e - P_a) \cdot A_e$$

(Eq. 2)

- F**thrust [N]
- I_{sp}**specific impulse [s]
- $\dot{m}$** mass flow rate [kg/s]
- P_e, P_a**exit / ambient pressure [Pa]

03

Experimental Results

3.1

KEY PERFORMANCE PARAMETERS

PEAK THRUST 312 N	BURN TIME 2.84 s	TOTAL IMPULSE 718 N·s
ISP (MEASURED) 163 s	AVG PC 4.2 MPa	BURN RATE (A) 8.26 mm/s · MPa⁻ⁿ

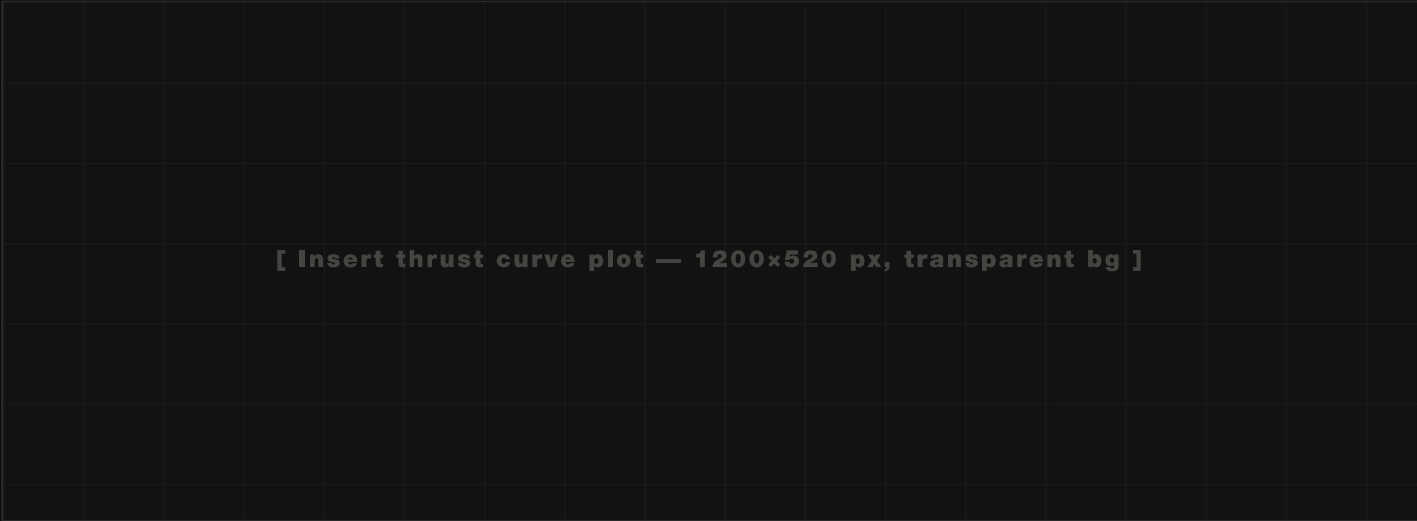
3.2

BURN RATE DATA

TEST ID	PRESSURE [MPa]	BURN RATE [mm/s]	Δ FROM MODEL [%]	STATUS
SB-01	2.10	14.3	+1.4	✓ Valid
SB-02	3.50	19.8	−0.8	✓ Valid
SB-03	5.00	26.1	+2.1	✓ Valid
SB-04	6.20	30.7	−3.6	⚠ Review

3.3

THRUST CURVE



[Hot-fire photograph — Test HF-03]

Fig. 1 VR-01 motor during hot-fire test HF-03. Visible plume length $\approx$ 380 mm at peak thrust. Photo: Vultur Rocketry, 2025.

04 Simulation vs. Experiment

OpenMotor was used to generate a predicted thrust curve using the empirically derived burn rate constants $a = 8.26$, $n = 0.32$. The table below summarizes the agreement between simulation and measured data.

PARAMETER	SIMULATED	MEASURED	ERROR [%]
Peak thrust [N]	307	312	1.6
Burn time [s]	2.90	2.84	2.1
Total impulse [N·s]	724	718	0.8
Isp [s]	165	163	1.2

05 Conclusions

The empirical burn rate law derived from strand burner data accurately predicts VR-01 motor performance to within 2.1% across all key parameters. The pressure exponent $n = 0.32$ confirms stable burning behavior within the operational pressure range (2.0–6.5 MPa).

Recommended next steps include extending the characterization to elevated temperatures (± 40 °C) and initiating nozzle erosion analysis for burn times exceeding 4 s.

06 References

